WEBRUM-99: Best Practice Summary

> **Series:** WEBRUM — Web Real User Monitoring | **Notebook:** 9 of 9 | **Created:** March 2026 | **Last Updated:** 04/25/2026

Overview

This notebook consolidates every actionable best practice from the WEBRUM series (notebooks 01-08) into a single definitive reference. Each practice specifies the exact setting, value, or action to take — no hedging, no "it depends." Practices are organized by category and prioritized as Critical, Recommended, or Optional.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
RUM Enabled	At least one web application with RUM injection active
Permissions	`storage:events:read`, `storage:metrics:read`, `storage:entities:read`
Previous Notebooks	WEBRUM-01 through WEBRUM-08 (for full context)

1. RUM Agent Configuration

#	Best Practice	Recommended Setting/Value	Priority	Source
1	Use automatic injection	Injection method: **Automatic** (OneAgent modifies HTML responses at the web server)	Critical	WEBRUM-01

#	Best Practice	Recommended Setting/Value	Priority	Source
2	Filter out bots from all RUM queries	Always apply `filter user.type == "REAL_USER"` in every DQL query against `user.sessions`	Critical	WEBRUM-01
3	Use manual injection for CDN/static-hosted apps	Add RUM `<script>` tag to `<head>` before all other scripts when no server-side OneAgent exists	Recommended	WEBRUM-02
4	Use agentless RUM only as last resort	Select agentless monitoring only when OneAgent cannot be deployed at all	Optional	WEBRUM-02
5	Set session timeout to 30 minutes	Keep the default 30-minute inactivity timeout for session boundaries	Recommended	WEBRUM-01
6	Enable Fetch API monitoring	**Settings > Web and mobile monitoring > Async web requests and SPAs > Fetch requests**: **Enabled**	Critical	WEBRUM-02
7	Deploy RUM + Synthetic together	Use synthetic for availability SLAs and baselines; use RUM for real user experience	Critical	WEBRUM-01, WEBRUM-08

2. SPA Instrumentation

#	Best Practice	Recommended Setting/Value	Priority	Source
8	Enable SPA route change capture	**Settings > Web and mobile monitoring > Async web requests and SPAs > SPA route changes**: **Enabled**	Critical	WEBRUM-02
9	Enable Async Web Requests capture	**Settings > Web and mobile monitoring > Async web requests and SPAs > Async web requests**: **Enabled**	Critical	WEBRUM-02
10	Configure user action naming rules for dynamic URLs	Replace dynamic segments: `/users/{id}` becomes `/users/{*}` via **Settings > Web and mobile monitoring > User action naming**	Critical	WEBRUM-02
11	Use `data-dtname` attribute on interactive elements	Add `data-dtname="Checkout"` to buttons/links for meaningful action names	Recommended	WEBRUM-02
12	Load RUM agent before the SPA router initializes	Place the RUM `<script>` tag in `<head>` before framework bootstrap scripts	Critical	WEBRUM-02
13	Use `dtrum.enterAction()` / `dtrum.leaveAction()` for custom actions	Wrap business-critical interactions in manual RUM actions when auto-detection is insufficient	Recommended	WEBRUM-02
14	Validate SPA instrumentation after setup	Query for `RouteChange` actions — if none appear, SPA monitoring is misconfigured	Critical	WEBRUM-02
15	Separate mobile WebView and desktop into distinct RUM apps	Create two Dynatrace applications; use application detection rules based on User-Agent string	Recommended	WEBRUM-02
16	Add custom User-Agent suffix for WebView detection	Android: `setUserAgentString()` / iOS: `applicationNameForUserAgent` with app name + version	Recommended	WEBRUM-02

#	Best Practice	Recommended Setting/Value	Priority	Source
17	Enable hybrid WebView monitoring in mobile SDK	Android: `hybridWebView { enabled true; domains "your-domain.com" }` / iOS: `DTXHybridApplication = true`	Recommended	WEBRUM-02
18	Never instrument third-party WebViews	Only add your own domains to `domains` / `DTXMonitoredDomains` — never OAuth, payment, or external pages	Critical	WEBRUM-02

3. Core Web Vitals Thresholds

#	Best Practice	Recommended Setting/Value	Priority	Source
19	Target LCP at or below 2.5 seconds	**Largest Contentful Paint p75 <= 2500ms**. Poor threshold: > 4000ms	Critical	WEBRUM-03
20	Target INP at or below 200 milliseconds	**Interaction to Next Paint p75 <= 200ms**. Poor threshold: > 500ms	Critical	WEBRUM-03
21	Target CLS at or below 0.1	**Cumulative Layout Shift p75 <= 0.1**. Poor threshold: > 0.25	Critical	WEBRUM-03
22	Use p75 (not average) for CWV evaluation	Google uses 75th percentile as the pass/fail threshold for CWV	Critical	WEBRUM-03
23	Track CWV per page, not just per app	Aggregate CWV by `action.name` to identify which specific pages fail thresholds	Recommended	WEBRUM-03
24	Set explicit `width` and `height` on all images	Prevents CLS from images loading without reserved space	Critical	WEBRUM-03
25	Use `font-display: swap` with fallback font metrics	Prevents CLS from FOUT (Flash of Unstyled Text) during web font loading	Recommended	WEBRUM-03
26	Use fixed dimensions on ads, embeds, and iframes	Set explicit `width` / `height` on all third-party content containers to prevent layout shifts	Recommended	WEBRUM-03
27	Break long JavaScript tasks to improve INP	Split tasks > 50ms using `requestIdleCallback`, `setTimeout(0)`, or web workers	Critical	WEBRUM-03
28	Batch DOM reads and writes to reduce INP	Avoid forced layout/reflow by separating DOM read and write operations	Recommended	WEBRUM-03
29	Build a CWV scorecard query	Use the single-query pattern from WEBRUM-03 to show % Good/NI/Poor for all three metrics	Recommended	WEBRUM-03
30	Track CWV trends at hourly (LCP/INP) and daily (CLS) granularity	Use `makeTimeseries` with `interval:1h` for LCP/INP, `interval:1d` for CLS	Recommended	WEBRUM-03

4. Performance Optimization

#	Best Practice	Recommended Setting/Value	Priority	Source
31	Target TTFB at or below 800 milliseconds	**Time to First Byte p75 <= 800ms** (Google recommendation)	Critical	WEBRUM-06

#	Best Practice	Recommended Setting/Value	Priority	Source
32	Use DNS prefetching for third-party domains	Add `<link rel="dns-prefetch" href="//cdn.example.com">` for external resources	Recommended	WEBRUM-06
33	Enable HTTP/2 on all web servers	Reduces TCP connection overhead; multiplexes requests on a single connection	Critical	WEBRUM-06
34	Enable TLS 1.3 with OCSP stapling	Reduces SSL handshake time by one round trip	Recommended	WEBRUM-06
35	Defer non-critical JavaScript	Use `defer` or `async` attributes on `<script>` tags that are not needed for initial render	Critical	WEBRUM-06
36	Lazy-load below-the-fold images	Use `loading="lazy"` on `<img>` tags not visible in the initial viewport	Recommended	WEBRUM-06
37	Monitor the DOM Interactive to Load Event gap	A large gap indicates excessive sub-resource loading; target < 2 seconds	Recommended	WEBRUM-06
38	Deploy CDN edge servers for high-TTFB regions	If TTFB is high for specific geographies, add CDN nodes closer to those users	Critical	WEBRUM-06
39	Prioritize slow pages by impact score	Calculate `impact_score = page_views * avg_duration_ms` and fix highest-impact pages first	Recommended	WEBRUM-06
40	Set page load performance SLA at 3 seconds	Track `% of page loads with duration <= 3s` as primary performance SLA	Critical	WEBRUM-08

5. Error Monitoring

#	Best Practice	Recommended Setting/Value	Priority	Source
41	Target error session rate below 2%	**% of sessions with error.count > 0 < 2%**	Critical	WEBRUM-05, WEBRUM-08
42	Rank errors by affected session count, not occurrence count	A retry loop in one session inflates occurrence count; `countDistinct(session.id)` is the true impact metric	Critical	WEBRUM-05
43	Monitor XHR/fetch errors separately from JS errors	Filter `error.type == "XHR_ERROR" or error.type == "FETCH_ERROR"` to isolate backend API failures	Recommended	WEBRUM-05
44	Enable rage click detection	Dynatrace captures rage clicks automatically; query `rage.click == true` to find frustrated users	Critical	WEBRUM-05
45	Correlate errors with session outcomes	Compare bounce rate and avg actions for "With Errors" vs "No Errors" sessions to quantify error impact	Recommended	WEBRUM-05
46	Segment error rates by browser family	Query `error.count > 0` grouped by `browser.family` to identify browser-specific issues	Recommended	WEBRUM-05
47	Alert on new error messages	Query errors with `min(timestamp)` in the last hour to detect errors introduced by recent deployments	Critical	WEBRUM-08

#	Best Practice	Recommended Setting/Value	Priority	Source
48	Use `dtrum.reportError()` for custom business errors	Report application-specific errors (e.g., payment validation failures) via the RUM API	Recommended	WEBRUM-05

6. Session Replay and Privacy

#	Best Practice	Recommended Setting/Value	Priority	Source
49	Set Session Replay default masking to "Mask user input"	**Settings > Session Replay > Data privacy > Default masking level**: **Mask user input**	Critical	WEBRUM-07
50	Block entire PII containers with `data-dtrum-block="true"`	Add `data-dtrum-block="true"` to any `<div>` containing sensitive data (SSN, CC, health info)	Critical	WEBRUM-07
51	Add CSS masking rules for all form inputs with sensitive data	**Settings > Session Replay > Data privacy**: add selectors like `.credit-card-form input`, `[data-sensitive]`	Critical	WEBRUM-07
52	Set replay sampling rate based on traffic volume	High-traffic production: **1-5%**; internal apps: **10-25%**; pre-prod: **50-100%**	Recommended	WEBRUM-07
53	Enable error-triggered replay capture	Configure conditional capture: automatically record sessions when JavaScript errors occur	Recommended	WEBRUM-07
54	Enable frustration-triggered replay capture	Configure conditional capture for sessions with rage clicks or exit intent	Recommended	WEBRUM-07
55	Err on over-masking, not under-masking	It is better to mask non-sensitive content than to accidentally capture PII	Critical	WEBRUM-07
56	Conduct a privacy review before enabling Session Replay in production	Review all captured content against GDPR, CCPA, and HIPAA requirements before go-live	Critical	WEBRUM-07
57	Use Dynatrace native replay over third-party tools	Native replay integrates with performance waterfall, backend traces, and Dynatrace Intelligence; third-party tools lack this correlation	Recommended	WEBRUM-07
58	Follow the replay investigation workflow	1) Identify via DQL, 2) Watch replay, 3) Correlate with waterfall, 4) Note pattern, 5) Quantify with DQL	Recommended	WEBRUM-07

7. Session Analysis

#	Best Practice	Recommended Setting/Value	Priority	Source
59	Define custom session properties for business context	Configure **Settings > Web and mobile monitoring > Session and user action properties** with: `session.user_id`, `session.plan_type`, `session.cart_value`, `session.ab_variant`	Critical	WEBRUM-04
60	Target bounce rate below 40%	**% of sessions with action.count == 1 < 40%**	Recommended	WEBRUM-04, WEBRUM-08
61	Segment sessions by engagement level	Bucket sessions: Bounce (1 action), Low (2-3), Medium (4-10), High (10+)	Recommended	WEBRUM-04

#	Best Practice	Recommended Setting/Value	Priority	Source
62	Track entry and exit pages	Use `takeFirst(action.name)` and `takeLast(action.name)` grouped by `session.id` to map user journeys	Recommended	WEBRUM-04
63	Use session properties for conversion tracking, not URL pattern matching	Mark conversions via session properties instead of fragile `action.name ~ "*confirmation*"` patterns	Critical	WEBRUM-04
64	Analyze sessions by hour of day for capacity planning	Group sessions by `getHour(timestamp)` over 7 days to identify peak traffic windows	Optional	WEBRUM-04
65	Segment by geography and device	Break down sessions by `country`, `browser.family`, `os.family`, and `connection.type` to find regional/device issues	Recommended	WEBRUM-04, WEBRUM-06

8. Dashboards and Alerting

#	Best Practice	Recommended Setting/Value	Priority	Source
66	Set Apdex threshold T = 3 seconds for page loads	Satisfied: <= 3s, Tolerating: 3-12s, Frustrated: > 12s	Critical	WEBRUM-08
67	Set Apdex threshold T = 2.5 seconds for XHR and route changes	Satisfied: <= 2.5s, Tolerating: 2.5-10s, Frustrated: > 10s	Critical	WEBRUM-08
68	Target Apdex score above 0.85	Apdex 0.85-0.93 = Good; 0.94-1.00 = Excellent; below 0.70 = investigate immediately	Critical	WEBRUM-08
69	Build an executive dashboard with 6 KPIs	Tiles: Apdex, session count, error rate %, bounce rate %, CWV pass rate %, avg session duration	Recommended	WEBRUM-08
70	Build an operational dashboard with real-time error view	Tiles: error count per 15m by type, active error table (last 1h), performance SLA % under 3s	Recommended	WEBRUM-08
71	Alert on error rate spike	Threshold: **> 5% of sessions with errors** sustained for **15 minutes**	Critical	WEBRUM-08
72	Alert on Apdex drop	Threshold: **Apdex < 0.7** sustained for **15 minutes**	Critical	WEBRUM-08
73	Alert on performance degradation	Threshold: **p75 page load > 5 seconds** sustained for **15 minutes**	Critical	WEBRUM-08
74	Alert on traffic anomaly	Threshold: **session count < 50% of previous day's hourly average**	Recommended	WEBRUM-08
75	Alert on CWV regression	Threshold: **LCP p75 > 4 seconds** sustained for **30 minutes**	Recommended	WEBRUM-08
76	Use metric event evaluation window of 15 minutes	**Settings > Anomaly detection > Metric events**: evaluation window **15m**, sliding window **5m**	Recommended	WEBRUM-08

#	Best Practice	Recommended Setting/Value	Priority	Source
77	Prefer Dynatrace Intelligence anomaly detection over static thresholds in production	Dynatrace Intelligence automatically baselines normal behavior, reducing false positives from seasonal patterns	Recommended	WEBRUM-08
78	Combine RUM and synthetic in a single dashboard	Use `append` to show side-by-side comparison: RUM real-user metrics vs synthetic clean-room metrics	Recommended	WEBRUM-08
79	Track CWV pass rate as executive KPI	Target: **> 75% of page loads meeting all three CWV thresholds**	Recommended	WEBRUM-08

9. DQL Query Patterns

#	Best Practice	Recommended Setting/Value	Priority	Source
80	Always include `from:` time range on RUM queries	Use `from:-1h` for real-time, `from:-24h` for daily, `from:-7d` for trends	Critical	WEBRUM-01
81	Always filter `userType == "REAL_USER"` on session queries	Excludes bots (`ROBOT`) and synthetic monitors (`SYNTHETIC`) from real-user analysis	Critical	WEBRUM-01
82	Use `isNotNull()` before aggregating optional fields	Apply `filter isNotNull(web_vitals.largest_contentful_paint)` before CWV aggregations	Critical	WEBRUM-03
83	Use `countDistinct(sessionId)` for error impact measurement	Never use `count()` alone — it inflates impact from retry loops in single sessions	Critical	WEBRUM-05
84	Require minimum sample size before drawing conclusions	Use `filter page_views > 20` (or similar) to avoid misleading statistics from low-volume pages	Recommended	WEBRUM-03, WEBRUM-06
85	Use nanosecond-to-millisecond conversion: `/ 1000000.0`	Dynatrace duration fields are in nanoseconds; divide by 1,000,000 for milliseconds, 1,000,000,000 for seconds	Critical	WEBRUM-03
86	Alias all aggregations before using in `sort`	Always write `summarize c = count()` then `sort c desc`, never `sort count() desc`	Critical	All

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